

Sharing energy in wider area: A two-layer energy sharing scheme for massive prosumers

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HIGHLIGHTS

- A novel two-layer energy sharing market is proposed, where massive prosumers can share in a wider area.
- It is theoretically proved that the marker equilibrium approaches to the social optimum in large-scale markets.
- A hierarchically distributed bidding algorithm without binary variables is proposed, avoiding the curse of dimensionality.

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ABSTRACT

The popularization of distributed energy resources transforms end-users from consumers into prosumers. Inspired by the sharing economy principle, energy sharing markets for prosumers are proposed to facilitate the utilization of renewable energy. This paper proposes a novel two-layer energy sharing market for massive prosumers, which can promote social efficiency by wider-area sharing. In this market, there is an upper-layer energy sharing market (U-ESM) in the distribution system and numerous lower-layer energy sharing markets (L-ESMs) in local communities. Prosumers in the same community share energy with each other in the L-ESM, which can be uncleared. The energy surplus and shortage of L-ESMs are cleared in the U-ESM. Thanks to the wider-area two-layer structure, the market outcome is near-social-optimal in large-scale systems. However, the proposed market forms a complex mathematical program with equilibrium constraints (MPEC). To solve the problem, we propose an efficient and hierarchically distributed bidding algorithm. The proposed two-layer market and bidding algorithm are verified on the IEEE 123-bus system with 11,250 prosumers, which demonstrates the practicality and efficiency for large-scale markets.

1. Introduction

1.1. Background

In distribution systems, the number of distributed energy resources, e.g., electric vehicles, distributed energy storages, and photovoltaic panels, is rapidly increasing [1]. The role of the end-user on the demand side has been transformed from a passive consumer into an active prosumer, which can produce or consume electricity [2]. Over decades, there have

been numerous researches on managing prosumers, for example, energy management [3], demand response [4], and micro-grids [5]. However, most works treat prosumers as controllable resources, ignoring that their interests are not identical to those of the distribution system operator.

Inspired by the sharing economy principle, energy sharing markets (ESMs) are proposed to provide prosumers with more incentives. It is inspired by the sharing economy principle that idle resources are shared to promote social efficiency. In the sharing market, energy producers

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benefit from selling excessive electricity, while energy consumers can purchase low-priced electricity. With the decline in flexibility and the increase in uncertainty on the supply side, prosumers on the demand side are playing an increasingly crucial role in power systems [6]. Compared to traditional participants in electricity markets (typically thermal power plants), prosumers have some distinct characteristics, e.g., massive quantity and variable roles as buyers or sellers [7]. Therefore, how to design market mechanisms that can accommodate flexible energy sharing among large-scale prosumers has become a significant research direction.

1.2. Related works

The initial form of energy sharing markets tends to simplify the prosumer model. Plenty of literature [8–10] regard the prosumer as a price-taker, where the market is commonly modeled as a Stackelberg game. A market operator who sets price acts as a leader, while prosumers make transactive decisions as followers according to the price. Ref. [8] proposes a sharing market among photovoltaic prosumers, where the market operator sets different prices for buying and selling. In Ref. [9], a day-ahead market among energy storage systems is formulated under the time-of-use price. The leader–follower model may harm the incentives of prosumers and leak their privacy. Numerous researchers turn to separate prosumers into pure sellers and buyers [11–13], where the double-sided auction mechanism is commonly utilized for clearing. In Ref. [11], a prosumer-generator double auction market with suppliers as intermediaries is proposed for joint real-time and forward markets. Ref. [12] proposes an adaptive bidding strategy to infer the supply and demand from historical trading records under a double auction market of energy storage systems. In Ref. [13], a day-ahead continuous double auction framework is proposed, which allows the trading of power and heat across all the upcoming timeslots. In the double-sided auction market, the flexibilities of prosumers are suppressed and network constraints are difficult to consider.

In recent years, a broader literature [14–16] concentrates on unlocking the flexibility in the sharing market. Prosumers are price-makers and they can freely choose to be a buyer or a seller according to the market clearing price. Ref. [14] uses the gains from trade to compare energy sharing benefits in different markets, showing the significance of the diversity of consumption profiles in trading. In Ref. [15], a novel energy sharing mechanism for prosumers is proposed, where the role of the prosumer is endogenously determined after the market clears. Ref. [16] develops a distributed peer-to-peer market, whose equilibrium is proved equivalent to the social optimum.

Energy sharing under network constraints has attracted much attention in recent years [17–19]. In theory, [20] proves that the generalized Nash equilibrium of markets under network constraints may be multiple or non-existent. Ref. [17] complements network constraints in the market model in [15] and then proposes a distributed generalized Nash equilibrium seeking algorithm. Ref. [18] further reveals that

some prosumers in the sharing market may strengthen their market power due to network congestion. In Ref. [19], a continuous double auction market in the low-voltage network is proposed with a sensitivity-based method. Inspired by the principles of distribution locational marginal pricing (DLMP) [21], this paper develops a pricing mechanism in the energy sharing market according to the violation of network constraints.

Most of the above works focus on small-scale energy sharing markets, whereas prosumers in distribution systems are becoming more and more fragmented and massive. As the number of prosumers participating in the sharing market explodes, the market clearings in the above works may suffer from incredible computational burdens. For instance, in cooperative-game-based markets, well-known methods like Shapley value [22], nucleolus [23], Nash bargaining [24] suffer from the curse of dimensionality as the number of prosumers increases. Although Ref. [25] adopts the K-means clustering method to improve the scalability of nucleolus calculation, it achieves this at the expense of inexactness. Stackelberg-game-based market models [8–10] are commonly transformed into mixed-integer programming (MIP) by the big-M method. As NP -hard problems, large-scale MIPs always suffer from the curse of dimensionality. Therefore, existing works are difficult to implement in the wider-area energy sharing over massive prosumers.

To efficiently share the energy of massive prosumers in a wider area, multi-layer energy sharing markets are formulated [26–28]. However, existing literature suffers from several limitations. Ref. [26] introduces integer variables to the market clearing problem, which limits its applicability to scenarios with large-scale prosumers. Ref. [27] neglects network constraints, failing to account for the impact of energy sharing on power flow limitations. Ref. [28] restricts prosumers to being price takers, which hampers their incentive to participate in energy sharing markets. To address these issues, this paper proposes a novel two-layer energy sharing market mechanism that effectively avoids the aforementioned problems. As shown in Table 1, we summarize the properties of the markets proposed in this paper and other related literature to highlight the superiority of the proposed market.

1.3. Contributions

The major contributions of this paper are two-fold:

1. **Scalable sharing scheme.** We propose a novel two-layer energy sharing market, which simultaneously accommodates massive prosumers and maintains network constraints. In the lower-layer energy sharing market (L-ESM), prosumers share energy with each other in the same community, where the energy may not be cleared. Then the uncleared energy is traded among L-ESM operators in the upper-layer energy sharing market (U-ESM), which achieves energy sharing in a wider area. Besides, we also prove that the market equilibrium is unique and approximate to the social optimum in large-scale markets.

Table 1
Comparison of this paper with other related literature.

References	Price-maker	Flexible role	Network-constrained	Integer-free	Large-scale
[8–10]	✗	✓	✓	✗	✗
[11–13]	✓	✗	✗	✓	✗
[14,15]	✓	✓	✗	✓	✓
[16]	✗	✓	✗	✓	✗
[17,18]	✓	✓	✓	✓	✗
[19]	✓	✗	✓	✓	✗
[22,23]	✗	✗	✗	✓	✗
[24,28]	✗	✗	✓	✓	✗
[25]	✗	✗	✗	✓	✓
[26]	✓	✓	✓	✗	✗
[27]	✓	✓	✗	✓	✗
This paper	✓	✓	✓	✓	✓

2. Efficient bidding algorithm. An efficient distributed bidding algorithm is proposed to clear the two-layer energy sharing market. The proposed energy sharing market problem belongs to a mathematical program with equilibrium constraints (MPEC). Existing works [8–10,29,30] commonly transform MPEC into MIP, which is NP -hard and inefficient in large-scale markets. Differently, we first reveal that the Nash equilibrium in the L-ESM equals to the solution to a convex optimization problem. Then we propose a hierarchically distributed bidding algorithm according to the monotonicity between the shared energy and price, avoiding introducing integer variables, which efficiently clears the whole market.

1.4. Organization

The rest of this paper is organized as follows. Section 2 establishes the two-layer energy sharing framework. The lower- and upper-layer ESMs are formulated in Sections 3 and 4, respectively. Section 5 verifies the effectiveness and efficiency of the proposed sharing market in a large-scale market. Section 6 concludes the paper.

2. Overall market architecture

Suppose that there is a set of local communities $\mathcal{N}$, indexed by $i \in \mathcal{N}$. In an arbitrary community i , a set of prosumers indexed by $j \in \mathcal{U}_i$ will share energy in an L-ESM i . Assume that every prosumer is organized by *one and only one* local community, i.e., $\mathcal{U}_i \cap \mathcal{U}_{i'} = \emptyset, \forall i, i' \in \mathcal{N}, i \neq i'$.

The two-layer architecture of the proposed sharing market is shown in Fig. 1. There is one L-ESM operator in each local community to achieve energy sharing among prosumers. Meanwhile, there is a U-ESM operator for energy sharing among all local communities. In the L-ESM, prosumers transact with each other within the same community, achieving local energy sharing. The sold and purchased energy in the L-ESM need not be perfectly balanced. The imbalance part is transacted by the L-ESM operator through energy sharing in the U-ESM, thus enabling large-scale prosumers to share energy across a wider area. In addition to ensuring the balance of sold and purchased energy, the U-ESM operator also considers the network congestion constraints caused by energy sharing.

The clearing method of the proposed market model is a distributed two-level iterative approach. The inner layer involves prosumer bidding managed by the L-ESM operator, and the outer layer is the L-ESM operator's bidding under the management of the U-ESM operator. When the iterative algorithm converges, the market clears, and energy is shared according to the results. This clearing algorithm can converge within seconds, making it suitable for real-time markets, which typically clears every 5–15 min.

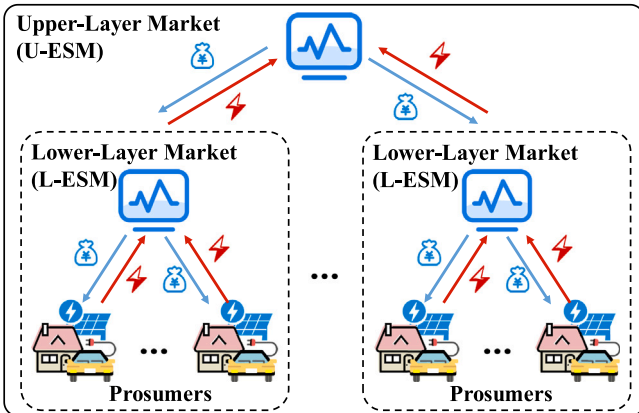


Fig. 1. Two-layer architecture of the energy sharing market.

Due to the uneven distribution of renewable energy production and load, the prosumers in the same community may be simultaneously willing to sell or buy electricity. Mandatory clearing in the L-ESM will damage prosumers' interests. Our model provides a new paradigm, where the uncleared energy can be shared in the U-ESM. Social efficiency is promoted since energy is more reasonably distributed.

3. Lower-layer energy sharing market

3.1. Energy sharing mechanism

The lower-layer energy sharing market of prosumers in the same local community is managed by the L-ESM operator. Assume that every prosumer has a distributed generator to generate electricity. Let p_j and D_j denote the power generation and the constant demand of prosumer j . The energy sharing market provides a platform for electricity transactions. The power balance equation for each prosumer j is

$$D_j + x_j = p_j, \quad (1)$$

where x_j is the shared energy amount. $x_j > 0$ if prosumer j is a seller, while $x_j < 0$ if it is a buyer.

Most studies follow similar prosumer models as (1). However, they neglect an important fact that prosumers can purchase energy from electric utilities in the main grid. In fact, it is necessary to consider the effect of the electric utility on prosumers, when studying the energy sharing market. As prosumer i can buy/sell electricity from/to the electric utility, the power balance Eq. (1) is replaced with

$$D_j + x_j + p_j^- = p_j + p_j^+, \quad (2)$$

where p_j^+ and p_j^- are the purchased and sold electricity, respectively.

The strategy of a prosumer is the amount of power generation and the transactions with the electric utility and the energy sharing market. The cost of power generation is commonly quadratic and independent of other prosumers. The buying/selling prices from/to the electric utility, denoted by ϖ^+ and ϖ^- , are fixed and identical for all prosumers. The energy sharing price in the L-ESM is decided by all prosumers in the same community, i.e., $\forall j \in \mathcal{U}_i$. Let $\varpi_i(\mathbf{x}^i)$ denote the energy sharing price in L-ESM i , where $\mathbf{x}^i$ is the compact form of the collection $\{x_j, \forall j \in \mathcal{U}_i\}$.

Therefore, the decision model of prosumer j can be formulated by the following optimization problem

$$\min_{p_j, p_j^+, p_j^-, x_j} \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- - \varpi_i(\mathbf{x}^i) x_j \quad (3a)$$

$$\text{s.t. } \underline{P}_j \leq p_j \leq \bar{P}_j, \quad : \quad \underline{\mu}_j, \bar{\mu}_j \quad (3b)$$

$$p_j^+ \geq 0, \quad p_j^- \geq 0, \quad : \quad \mu_j^+, \mu_j^- \quad (3c)$$

$$D_j + x_j + p_j^- = p_j + p_j^+, \quad : \quad \mu_j^{PB}, \quad (3d)$$

where $c_j > 0$ and b_j are constant parameters of the power generation cost, $\underline{P}_j$ and $\bar{P}_j$ are the lower and upper bounds of power generation with $\underline{P}_j \leq \bar{P}_j$, and $\underline{\mu}_j, \bar{\mu}_j, \mu_j^+, \mu_j^-$, and μ_j^{PB} are Lagrangian multipliers with respect to these constraints.

Remark 1. (Requirements for prosumers) During the implementation of the proposed market, a significant obstacle that may be encountered is the heterogeneity of prosumers. To develop the proposed energy sharing market, prosumers are required to install smart meters or microprocessor-based devices that can execute the following three types of operations. Firstly, prosumers must possess metering capabilities to acquire their own parameters, e.g., generation regions and load profiles. Secondly, prosumers should be able to establish bidirectional communication channels with the market operator to receive pricing signals and upload energy sharing data. Lastly, prosumers need to be capable of solving small-scale linearly constrained quadratic programming problems to optimize their energy sharing strategies in response to dynamic electricity prices.

The energy sharing price $\varpi_i(x^i)$ is defined as [29]

$$\varpi_i(x^i) = \varpi_i^0 - a_i \sum_{j \in \mathcal{U}_i} x_j, \quad (4)$$

where ϖ_i^0 is the base price and $a_i > 0$ is the price elasticity. $\sum_{j \in \mathcal{U}_i} x_j$ indicates the supply–demand relationship.

The energy-sharing clearing price $\varpi_i(x^i)$ is linearly adjusted based on the base price ϖ_i^0 according to the energy supply–demand relationship $\sum_{j \in \mathcal{U}_i} x_j$. When supply exceeds demand, i.e., $\sum_{j \in \mathcal{U}_i} x_j > 0$, the clearing price drops; otherwise, it rises. This pricing model is highly flexible and elastic. It can capture energy supply–demand fluctuations and guide prosumers to adjust their trading strategies accordingly. Therefore, the energy supply and demand do not necessarily have to be balanced, i.e., $\sum_{j \in \mathcal{U}_i} x_j$ is not necessarily equal to zero. L-ESM operators can share the unbalanced energy in U-ESM. As discussed in Section 4, the sharing price for L-ESM operators in U-ESM can be different from $\varpi_i(x^i)$, the energy-sharing price among prosumers. The L-ESM operators can obtain profits through the price difference in the two-layer energy sharing market.

Different from the L-ESM, the buying/selling price from/to the electric utility is fixed. Electric utilities are more inclined to sell electricity rather than buy it since bought energy fluctuates and increases the operation burden. Selling electricity is a mandatory responsibility for electric utilities, while buying from prosumers is not necessary. Besides, the electric utility always plays the role of a price-maker due to the supply–demand relationship. Thus electric utilities intend to arbitrage from electricity sellers and buyers by setting different buying and selling prices. It is widely adopted in electricity markets [31,32]. In economics, this phenomenon is called price discrimination.

Assumption 1 (Price discrimination). The prices of the electricity utility satisfy $\varpi^+ > \varpi^- > 0$.

In the proposed sharing market, price discrimination prevents prosumers from arbitraging from the electric utility. A prosumer will either buy electricity from the electric utility or sell to it, but not both. See the following proposition.

Proposition 1. Under Assumption 1, given $x_{j'}, \forall j' \in \mathcal{U}_i \setminus \{j\}$, the optimal purchased and sold electricity are complementary, i.e., $p_j^+ p_j^- = 0$.

The proof of Proposition 1 can be found in Appendix A.

3.2. Energy sharing as a nash game

The prosumer model (3) constitutes a Nash game, where prosumers' payoffs depend on others in the same community. Denote by s_j the collection of the decision variables p_j, p_j^+, p_j^-, x_j of prosumer j . Let s_{-j} denote the compact form of prosumers in community set $\mathcal{U}_i$ except j , i.e., $\mathcal{U}_i \setminus \{j\}$. The payoff and action set of prosumer j can be denoted by $F_j(s_j, s_{-j})$ and S_j , respectively. The Nash game model (3) can be simplified by

$$\min_{s_j \in S_j} F_j(s_j, s_{-j}). \quad (5)$$

The Nash game is composed of the following three elements:

1. The set of players $\mathcal{U}_i$;
2. Action sets $S_j, \forall j \in \mathcal{U}_i$, and the strategy space $S^i := \prod_{j \in \mathcal{U}_i} S_j$;
3. Payoffs $F_j(s_j, s_{-j}), \forall j \in \mathcal{U}_i$.

Then we will characterize some properties of the equilibrium of the L-ESM.

Definition 1 (Nash equilibrium). A strategy profile $s^i := \{s_j, \forall j \in \mathcal{U}_i\}$ is a Nash equilibrium (NE) of the Nash game (5), if

$$F_j(s_j, s_{-j}) \leq F_j(s_j, \hat{s}_{-j}), \forall s_j \in S_j, j \in \mathcal{U}_i. \quad (6)$$

In this paper, let $s^{i*} = \{p_j^*, p_j^{+*}, p_j^{-*}, x_j^*, \forall j \in \mathcal{U}_i\}$ denote an NE of the Nash game (5) in L-ESM i . The following proposition proves the uniqueness of the NE.

Proposition 2. Under Assumption 1, the Nash game (5) exists a unique NE s^{i*} , which is also the unique optimal solution to the following problem.

$$\begin{aligned} \min_{p_j, p_j^+, p_j^-, x_j} \sum_{j \in \mathcal{U}_i} \left\{ \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- - \varpi_i^0 x_j \right\} \\ + \frac{a_i}{2} \left(\sum_{j \in \mathcal{U}_i} x_j \right)^2 + \frac{a_i}{2} \sum_{j \in \mathcal{U}_i} x_j^2 \end{aligned} \quad (7a)$$

$$s.t. \quad \underline{p}_j \leq p_j \leq \bar{p}_j : \underline{\mu}_j, \bar{\mu}_j, \forall j \in \mathcal{U}_i \quad (7b)$$

$$p_j^+ \geq 0, \quad p_j^- \geq 0 : \mu_j^+, \mu_j^-, \forall j \in \mathcal{U}_i \quad (7c)$$

$$D_j + x_j + p_j^- = p_j + p_j^+ : \mu_j^{PB}, \forall j \in \mathcal{U}_i. \quad (7d)$$

The proof of Proposition 2 can be found in Appendix B. Proposition 2 indicates that the Nash game of the L-ESM is well defined.

Then we prove that prosumers are willing to participate in the L-ESM to share energy. Let F_j^{-x} denote the optimal cost of prosumer j that only trades with the electric utility.

Proposition 3. Under Assumption 1, we have for $\forall j \in \mathcal{U}_i$

$$F_j(s_j^*, s_{-j}^*) \leq F_j^{-x}. \quad (8)$$

Proposition 3 holds directly since F_j^{-x} is the optimal value of problem (3) with an extra constraint $x_j = 0$.

3.3. Bidding algorithm

The bidding algorithm of the L-ESM is presented in Algorithm 1. Prosumers only need to submit the decision of shared energy to the market. Their private information, e.g., the energy demand or the generation parameters, will not be leaked to the market operator or other prosumers. Prosumer's problem (9a) is equivalent to problem (3) with $x_{j'} \equiv x_{j'}(h-1), \forall j' \in \mathcal{U}_i \setminus \{j\}$. Note that each prosumer does not have to communicate with others. It can directly deduce $\sum_{j' \in \mathcal{U}_i \setminus \{j\}} x_{j'}$ by $\varpi_i(h-1)$. After solving (9a), the real bidding is averaged by the last one as (9b), which describes prosumers' inertia in the decision. From the perspective of the algorithm, the step size ρ_i decides the convergence of the bidding.

Algorithm 1 Bidding Algorithm of L-ESM i

Input: Error tolerance ε_i , base price ϖ_i^0 , price elasticity a_i , initial energy sharing price $\varpi_i(0)$, initial shared energy $x_j(0), \forall j \in \mathcal{U}_i$, step size $\rho_i \in (0, 1)$, and iteration index $h = 1$.

Output: Prosumer strategy $p_j^*, p_j^{+*}, p_j^{-*}, x_j^*, \forall j \in \mathcal{U}_i$ and energy sharing price $\varpi_i(x^{i*})$.

S1: Each prosumer $j \in \mathcal{U}_i$ decides a sharing strategy $s_j(h) = \{p_j(h), p_j^+(h), p_j^-(h), x_j(h)\}$ with $\varpi_i(h-1)$ by solving the following problem

$$\begin{aligned} \tilde{s}_j(h) = \arg \min_{s_j \in S_j} \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- \\ - [\varpi_i(h-1) + a_i x_j(h-1) - a_i x_j] x_j \end{aligned} \quad (9a)$$

$$s_j(h) = \rho_i \tilde{s}_j(h) + (1 - \rho_i) s_j(h-1) \quad (9b)$$

and then send the shared energy $x_j(h)$ to the L-ESM operator.

S2: After receiving all $x_j(h), j \in \mathcal{U}_i$, the L-ESM operator readjusts the energy sharing price by

$$\varpi_i(h) = \varpi_i^0 - a_i \sum_{j \in \mathcal{U}_i} x_j(h) \quad (10)$$

and then broadcasts it to all prosumers.

S3: If $|\varpi_i(h) - \varpi_i(h-1)| \leq \varepsilon_i$, terminate the bidding and output the latest strategy and price; otherwise, set $h \leftarrow h+1$ and go to S1.

3.4. Prices in equilibrium

Existing works commonly assume that prosumers can only trade in the energy market. However, they ignore the influence of electric utilities on sharing markets. We will analyze the condition where prosumers can share energy with each other in the sharing market or/and trade with the electric utility. The interaction among the sharing price $\varpi_i(x^i)$, the buying/selling prices ϖ^+ , ϖ^- , and the shadow prices of prosumers are revealed. To this end, we present the definition of shadow price quantifying the marginal consumption/production price of the prosumer.

Definition 2 (Shadow price). The shadow price of prosumer j , μ_j^{PB*} , is defined as the optimal Lagrangian multiplier μ_j^{PB} with respect to the power balance constraint (3d) at the NE.

The relationship among the shadow price, the electric utility price, and the energy sharing price is clarified below.

Proposition 4. Under Assumption 1, at the NE $s^{i*} = \{p_j^*, p_j^{+*}, p_j^{-*}, x_j^*, \forall j \in \mathcal{U}_i\}$, we have

1. the shadow price of each prosumer $j \in \mathcal{U}_i$ satisfies

$$\varpi^- \leq \mu_j^{PB*} \leq \varpi^+. \quad (11)$$

2. the shared energy is given by

$$x_j^* = \frac{\varpi_i(x^{i*}) - \mu_j^{PB*}}{a_i}. \quad (12)$$

3. the energy sharing price at equilibrium satisfies

$$\varpi_i(x^{i*}) = \frac{\varpi_i^0 + \sum_{j \in \mathcal{U}_i} \mu_j^{PB*}}{1 + |\mathcal{U}_i|}. \quad (13)$$

4. if the base price ϖ_i^0 in the L-ESM (4) satisfies

$$\varpi^- \leq \varpi_i^0 \leq \varpi^+, \quad (14)$$

then we have

$$\varpi^- \leq \varpi_i(x^{i*}) \leq \varpi^+. \quad (15)$$

Besides, even without condition (14), the result (15) still holds if the number of prosumers is sufficiently large, i.e.,

$$\varpi^- \leq \varpi_i(x^{i*}) \leq \varpi^+, |\mathcal{U}_i| \rightarrow \infty. \quad (16)$$

The proof of Proposition 4 can be found in Appendix C. Proposition 4 reveals the competitiveness of the L-ESM against the electric utility. The energy sharing price $\varpi_i(x^{i*})$ are commonly located in the range $[\varpi^-, \varpi^+]$. It means that prosumers can buy electricity at a lower price than ϖ^+ and sell at a higher price than ϖ^- . That is the real reason why prosumers are willing to participate in the L-ESM. Besides, from the perspective of the market, the interest of the electric utility by price discrimination is cut down due to the competition of the L-ESM. Because of the existence of the sharing market, the electric utility cannot severely set discriminative prices.

Eq. (12) indicates that the role of a prosumer depends on the values of the energy sharing price and the shadow price. If the energy sharing price $\varpi_i(x^{i*})$ is larger than the shadow price μ_j^{PB*} , the prosumer will sell energy, i.e., $x_j^* > 0$. Otherwise, the prosumer will be an energy buyer.

Fig. 2 illustrates the relationship among prices. The shadow price of a prosumer guarantees its role in the transaction. Prosumers with a lower shadow price tend to sell electricity to those with a higher shadow price. The real energy sharing price $\varpi_i(x^{i*})$ is the average of these sharing prices with the base price.

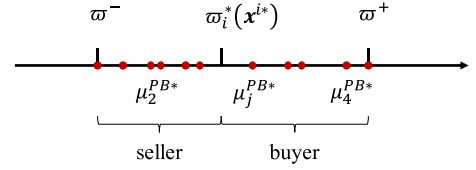


Fig. 2. An illustration of the relationship among prices.

4. Upper-layer energy sharing market

4.1. Energy sharing mechanism

Recall that in each L-ESM, the shared energy does not have to be cleared. These L-ESM operators will share the uncleared energy in the U-ESM managed by the U-ESM operator. In the U-ESM, an L-ESM operator can be regarded as an aggregated “prosumer”, whose uncleared energy is the shared electricity.

In the U-ESM, the energy shared by L-ESMs is traded at the base price ϖ_i^0 . The U-ESM operator can adjust the base price to influence the equilibrium of L-ESMs. Let $x_j^*(\varpi_i^0)$ denote the shared energy of prosumer j at the NE under the base price ϖ_i^0 . Define the uncleared energy of community i , i.e., the shared energy in the U-ESM, as

$$y_i(\varpi_i^0) = \sum_{j \in \mathcal{U}_i} x_j^*(\varpi_i^0). \quad (17)$$

Eq. (17) aggregates the volume–price curve of an L-ESM.

The clearing rule of the U-ESM is solving the following problem to decide base prices in all L-ESMs

$$\min_{\varpi^0} f(\varpi^0) \quad (18a)$$

$$\text{s.t. } \varpi^0 := \{\varpi_i^0, i \in \mathcal{N}\}$$

$$\sum_{i \in \mathcal{N}} y_i(\varpi_i^0) = 0 \quad (18b)$$

$$\sum_{i \in \mathcal{N}} \pi_{il} y_i(\varpi_i^0) \leq F_l, \forall l \in \mathcal{L}, \quad (18c)$$

where $\mathcal{L}$ is the set of network constraints, π_{il} is the parameter quantifying the factor of L-ESM i to network constraint l , F_l is the constant physical limitation of network constraint l . Constraint (18b) is the energy clearing limitation, while constraint (18c) guarantees the power flow security under network constraints. In this paper, the classic DC power flow model [33] is applied to describe the network congestion constraints. It should be noted that any linearized power flow model can be utilized in the proposed energy sharing mechanism.

The objective $f(\varpi^0)$ can be some function of social welfare. However, purely attaining a feasible solution to the problem is extremely hard. Noting that every $y_i(\varpi_i^0)$ implies an NE of an L-ESM, problem (18) belongs to an MPEC. To solve it, integer variables are commonly introduced to simplify complementary slackness conditions in equilibriums, which transforms the MPEC to a mixed-integer program (MIP) [8–10, 29,30]. There are massive prosumers in the energy sharing problem. An enormous number of integer variables are required for transforming complementary slackness conditions (C.5)–(C.8), which leads to the curse of dimensionality in solving the MIP problem.

To this end, we propose a novel bidding algorithm for the U-ESM clearing problem. Then we prove that the algorithm can attain a feasible price strategy, which simultaneously approaches to the social optimal.

Algorithm 2 Bidding Algorithm of the U-ESM

Input: Error tolerance ε , initial energy sharing price $\varpi^{PB}(0)$, initial network congestion prices $\varpi_l(0), \forall l \in \mathcal{L}$, initial base price $\varpi_i^0(0) = \varpi^{PB}(0) + \sum_{l \in \mathcal{L}} \pi_{il} \varpi_l(0), \forall i \in \mathcal{N}$, step sizes $\alpha^{PB}, \alpha_l > 0$, and iteration index $k = 0$.

Output: Shared energy $y_i^*, \forall i \in \mathcal{N}$ and clearing prices $\varpi^{PB*}, \varpi_i^{NET*}, \varpi_i^{0*}, \forall i \in \mathcal{N}$.

S1: Given the base price $\varpi_i^0(k)$, each L-ESM $i \in \mathcal{N}$ clears by **Algorithm 1** and then sends the shared energy $y_i(\varpi_i^0(k))$ to the U-ESM operator.

S2: After receiving all $y_i(\varpi_i^0(k)), i \in \mathcal{N}$, the U-ESM operator readjusts the energy sharing price by (20) and then broadcasts prices $\varpi_i^0(k+1)$ to all L-ESMs.

S3: If $|\varpi_i(k) - \varpi_i(k-1)| \leq \varepsilon, \forall i \in \mathcal{N}$, terminate the bidding and output the latest strategy and prices; otherwise, set $k \leftarrow k+1$ and go to **S1**.

4.2. Bidding algorithm

The basic idea is to linearly adjust the sharing price according to the violation level of the power balance and network congestion constraints. To this end, we first analyze the property of $y_i(\varpi_i^0)$, yielding the following proposition.

Proposition 5. Under **Assumption 1**, the uncleared energy $y_i(\varpi_i^0)$ in each L-ESM $i \in \mathcal{N}$ is monotonically non-decreasing with respect to ϖ_i^0 .

The proof of **Proposition 5** can be found in **Appendix D**. Based on this, we define the base price as

$$\varpi_i^0 = \varpi^{PB} + \varpi_i^{NET} = \varpi^{PB} + \sum_{l \in \mathcal{L}} \pi_{il} \varpi_l, \quad (19)$$

where ϖ^{PB} is the basic sharing price based on the power balance, ϖ_i^{NET} is the additional price on L-ESM i according to the network congestions, ϖ_l is the congestion price of network constraint l , and parameter π_{il} quantifies the sensitivity of energy sharing of L-ESM i on network constraint l . The energy sharing between L-ESM operators uses ϖ^{PB} as the actual transaction price, while ϖ_i^{NET} is the unit price of the additional charge from network congestion. When network constraints are violated, the additional fee is charged to guide L-ESM's energy sharing decisions; otherwise, the additional charge is zero. For a congested transmission line, the U-ESM will update the corresponding congestion price ϖ_l . Then different fees are charged to each L-ESM operator based on their power flow distribution factors π_{il} .

Note that the basic sharing price ϖ^{PB} reflects the state of power balance. If the power is deficit, according to the monotonicity of $y_i(\varpi_i^0)$ in **Proposition 5**, the price ϖ^{PB} can be higher to encourage power production and discourage power consumption; otherwise, the price ϖ^{PB} should be reduced. The network congestion price ϖ_l also has the same property. Therefore, we design the update model of the prices as

$$\varpi^{PB}(k+1) = \varpi^{PB}(k) - \alpha^{PB} \sum_{i \in \mathcal{N}} y_i(\varpi_i^0(k)) \quad (20a)$$

$$\varpi_l(k+1) = \left[\varpi_l(k) - \alpha_l \left(\sum_{i \in \mathcal{N}} \pi_{il} y_i(\varpi_i^0(k)) - F_l \right) \right]^- \quad (20b)$$

$$\varpi_i^0(k+1) = \varpi^{PB}(k+1) + \sum_{l \in \mathcal{L}} \pi_{il} \varpi_l(k+1), \quad (20c)$$

where $[\cdot]^-$ is the projection onto the range $(-\infty, 0]$, $\alpha_l > 0, \forall l \in \mathcal{L}$ are step sizes. The complete bidding algorithm is presented in **Algorithm 2**.

4.3. Social efficiency

Then we will prove the convergence of **Algorithm 2** and reveal its market efficiency.

Proposition 6. Under **Assumption 1**, the bidding **Algorithm 2** converges to a unique equilibrium $\{p_j^*, p_j^{+*}, p_j^{-*}, x_j^*, y_i^*, \forall j \in \mathcal{U}, i \in \mathcal{N}\}$, which is also the unique optimal solution to the following problem.

$$\min_{p_j, p_j^+, p_j^-, x_j, y_i} \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{U}_i} \left\{ \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- \right\} + \sum_{i \in \mathcal{N}} \left\{ \frac{a_i}{2} y_i^2 + \frac{a_i}{2} \sum_{j \in \mathcal{U}_i} x_j^2 \right\} \quad (21a)$$

$$\text{s.t. } \{p_j, p_j^+, p_j^-, x_j\} \in S_j, \forall j \in \mathcal{U}, i \in \mathcal{N} \quad (21b)$$

$$y_i = \sum_{j \in \mathcal{U}_i} x_j, \forall i \in \mathcal{N} \quad (21c)$$

$$\sum_{i \in \mathcal{N}} y_i = 0, \quad : \quad \lambda^{PB} \quad (21d)$$

$$\sum_{i \in \mathcal{N}} \pi_{il} y_i \leq F_l, \quad : \quad \lambda_l, \forall l \in \mathcal{L} \quad (21e)$$

The proof of **Proposition 6** can be found in **Appendix E**. **Proposition 6** directly shows that the market clear result by **Algorithm 2** is a feasible solution to problem (18). As a result, the base price satisfies $\varpi_i^0 = -\lambda^{PB} - \sum_{l \in \mathcal{L}} \pi_{il} \lambda_l$, which can be regarded as the DLMP [21] and may integrate existing DLMP approaches.

Then we will show that the market clear result also performs excellently from the perspective of social welfare. Define the social optimum of the whole system as follows.

Definition 3 (Social optimum). A strategy of prosumers and L-ESM operators $\{p_j, p_j^+, p_j^-, x_j, y_i, j \in \mathcal{U}, i \in \mathcal{N}\}$ is socially optimal if it is the solution to the following problem

$$\min_{p_j, p_j^+, p_j^-, x_j, y_i} \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{U}_i} \left\{ \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- \right\} \quad (22a)$$

$$\text{s.t. } (21b) - (21e). \quad (22b)$$

Remark 2. (Social efficiency) Note that the only difference between problems (21) and (22) is the quadratic terms in the objective function (21a). The terms can be regarded as the social welfare loss from market competition, whose degree depends on the price elasticities $a_i, \forall i \in \mathcal{N}$. In the case study in **Section 5.3**, we show that the price elasticities are supposed to decrease with the increasing scale of prosumers. Therefore, it reveals that the market clearing result approximates to the social optimum in large-scale markets.

5. Case study**5.1. Setup**

In this section, simulation experiments are carried out in MATLAB 2021 A on a desktop with an Intel i7-10700 CPU and 16 GB memory. As shown in **Fig. 3**, the IEEE 123-bus system is adopted, where every bus is integrated with an L-ESM. The number of prosumers in an L-ESM is in the range of [50, 525], depending on the raw data of active power at the bus. The total number of L-ESMs and prosumers are up to 123 and 11250, respectively. The capacities of distribution lines {10, 63}, {35, 36}, {4, 74}, {8, 9}, {15, 16}, {78, 79}, and {95, 96} are 0.5, 12, 1.5, 21, 9, 10, and 15 MW, respectively.

The buying and selling prices ϖ^+, ϖ^- of the electric utility are set to 0.2\$/kW and 0.05\$/kW, respectively. For each L-ESM, the price elasticity a_i is randomly chosen within the range $[2.5, 5] \times 10^{-3} / |\mathcal{U}_i|$ \$/kW. The parameters of prosumers are randomly chosen within the ranges: $c_j \in [0.5, 1] \times 10^{-3}$ \$/kW², $b_j \in [0.01, 0.05]$ \$/kW, $D_j \in [0, 40]$ kW, and $\underline{P}_j = 0$ kW. Since renewable energy is fluctuating, we design five types of maximal power generations randomly chosen within the ranges [35, 50] kW, [20, 35] kW, [15, 25] kW, [5, 10] kW, and [0, 5] kW,

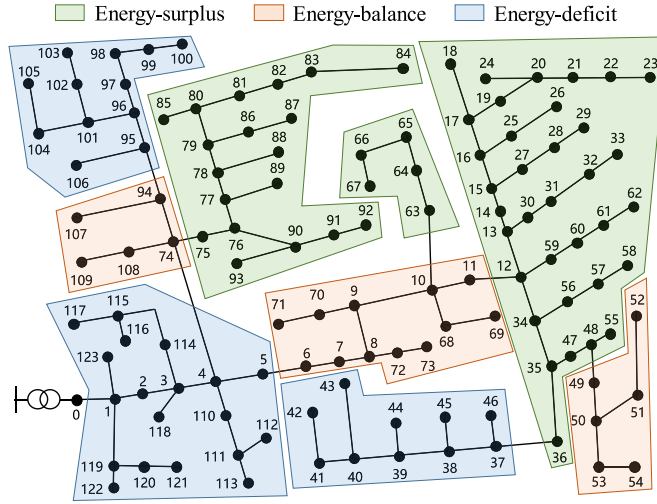


Fig. 3. Configuration of the IEEE 123-bus system.

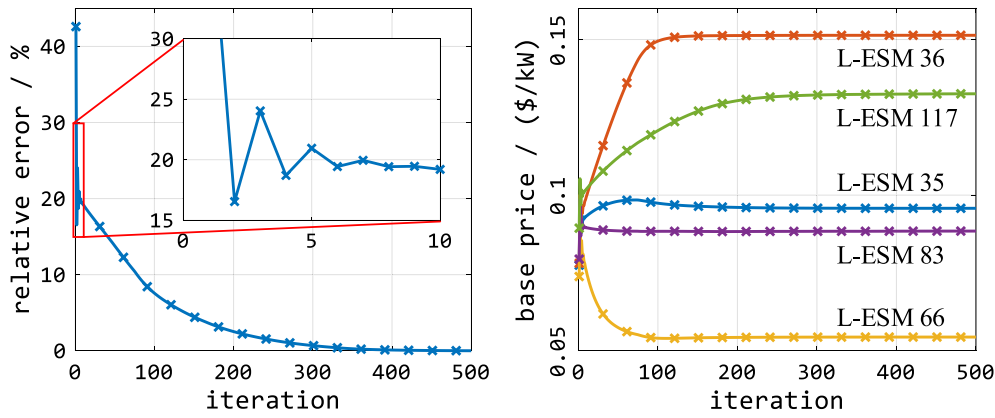


Fig. 4. Total cost relative error and sharing price during iterations in the U-ESM.

respectively. Then the L-ESMs are divided into three types: energy-surplus/balance/deficit communities. In energy-surplus L-ESMs, 80 % and 20 % prosumers pick the maximal power generations from the first and second ranges, respectively. The maximal power generations of prosumers in energy-balance L-ESMs uniformly fall into the five ranges. In energy-deficit ones, 80 % and 20 % prosumers get their maximal power generations from the fifth and fourth ranges, respectively. Set the step sizes $\alpha^{PB} = 1 \times 10^{-6} \$/kW$, $\alpha_l = 5 \times 10^{-7} \$/kW$, $\forall l \in \mathcal{L}$ in (20). For Algorithm 1, we design an adaptive step size with the initial value $\rho_i = 0.2$, $\forall i \in \mathcal{N}$. During iterations, halve ρ_i if $\varpi_i(h) - \varpi_i(h-1) > 10^{-3} \$ \wedge \varpi_i(h) - \varpi_i(h+1) > 10^{-3} \$$ or $\varpi_i(h-1) - \varpi_i(h) > 10^{-3} \$ \wedge \varpi_i(h+1) - \varpi_i(h) > 10^{-3} \$$.

Remark 3. (Issues of dataset) Although the dataset used in the case study is not derived from the real world, our simulation results maintain strong reliability. On the one hand, given the current lack of large-scale prosumer datasets, we adopted randomized sampling within parameter ranges from Ref. [15]. This approach ensures empirical compliance with the real world and aligns with methodologies widely adopted in related literature. On the other hand, the properties of the proposed energy sharing market mechanism have been rigorously proven, demonstrating theoretically robust effects against variant datasets. Consequently, the representativeness and reliability of simulation results are well substantiated.

5.2. Main results

The U-ESM clearing process by Algorithm 2 is shown in Fig. 4. We calculate the sum of all 11,250 prosumers' costs during 500 iterations and record its relative error from the exact value by solving problem (6) in the left subfigure. It verifies the convergence of Algorithm 2 and the correctness of Proposition 6. The computational time of Algorithm 2 and solving problem (6) is 10.2 s and 23.8 s, respectively. Since the biddings of all prosumers in Algorithm 2 are serially calculated, the average time spent by one prosumer is only 9.07×10^{-4} s, 1/26,193 times of solving problem (6). It validates the scalability of the proposed market clearing algorithm in large-scale markets. We pick up five L-ESMs (35, 36, 66, 83, and 117) to display their base prices ϖ_i^0 during iterations in the right subfigure. Due to the congestion of distribution line {35,36}, the base price of L-ESM 36 is 0.056 \$/kW larger than that of L-ESM 35.

The market clearing process in the L-ESMs 35, 36, 66, 83, and 117 by Algorithm 1 during the 25th iteration of the U-ESM bidding is displayed in Fig. 5. In the left subfigure, the total costs of prosumers in one community are calculated and compared to the value obtained by solving problem (7), which verifies the convergence of Algorithm 1 and the correctness of Proposition 2. The right subfigure shows the sharing prices during iterations. The required iteration numbers of L-ESMs to reach the error tolerance 10^{-8} are different. The average number of iterations over the whole algorithm is just 15.1, which is surprisingly quick.

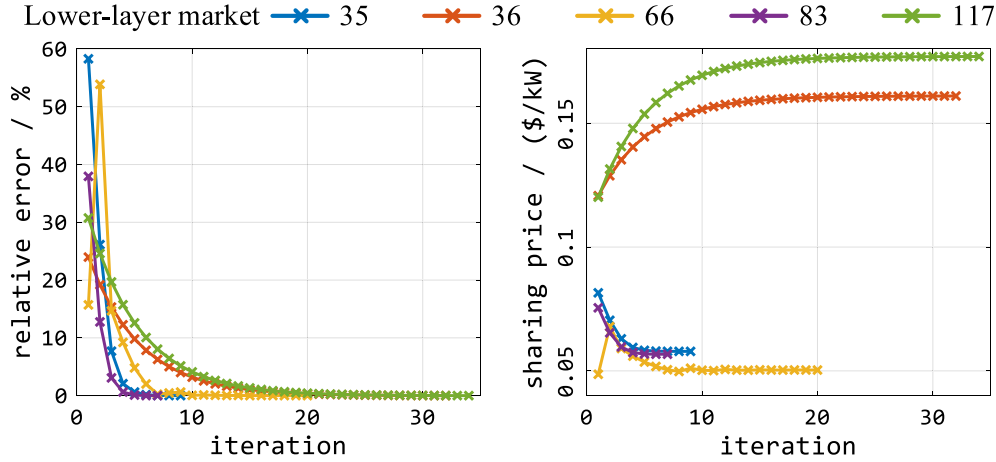


Fig. 5. Total cost relative errors and sharing prices during iterations in the L-ESMs.

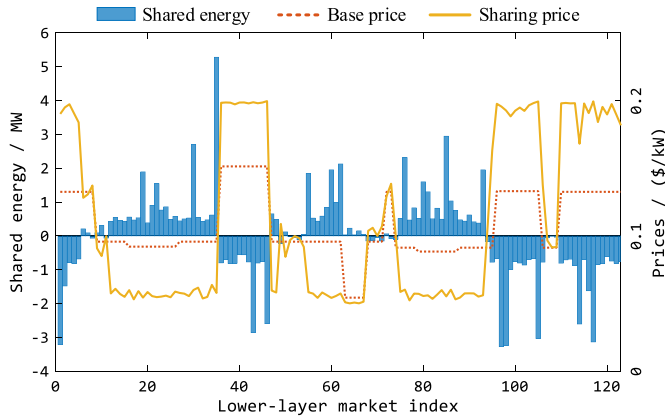


Fig. 6. Upper-layer energy sharing market outcome.

Fig. 6 shows the U-ESM outcome. The L-ESMs vary in the base price due to distribution line congestion. For instance, the base prices of L-ESMs 63–67 are near to the selling price $\varpi^- = 0.05$ \$/kW since the shared energy reaches the capacity of distribution line {10, 63}. Similarly, the base price of L-ESMs 36–46 are extremely high because of the congestion of line {35, 36}. This pricing mechanism can indirectly guide the energy sharing by influencing the base prices. The figure also shows that the final sharing prices are all located within the range of $[\varpi^-, \varpi^+]$, verifying Proposition 2. It reveals the competitiveness of the proposed sharing market against the electric utility: higher selling price and lower purchasing price.

5.3. Social efficiency

We consider five conditions for energy sharing:

1. **Self-sufficiency (SS)**, energy balance by itself without energy sharing, i.e., setting the value of x_j to be fixed at 0 in prosumer's decision model (3);
2. **Local-community sharing (LS)**, sharing and clearing energy with prosumers in the same community, i.e., choosing appropriate value of base price ϖ_i^0 such that $y_i = 0$;
3. **Local-community optimization (LO)**, local-community cooperation for social optimum, i.e., solving the social optimization problem (22) with $y_i = 0, \forall i \in \mathcal{N}$;
4. **Global sharing (GS)**, the proposed two-layer market, i.e., implementing Algorithm 2;

Table 2

Total cost in different conditions.

Conditions	SS	LS	LO	GS	GO
Total cost/k\$	19.95	16.90	16.90	11.32	9.08

5. **Global optimization (GO)**, global cooperation for social optimum i.e., solving the social optimization problem (22).

Table 2 lists the total costs of all prosumers in the five conditions. From SS to LS to GS, with the expansion of the sharing domain, the total cost remarkably reduces, highlighting the significance of global sharing. The local-community sharing (LS) almost achieves the same total cost as that of local-community optimization (LO), i.e., the theoretical lower bound. However, due to the uneven distribution of energy resources, there is still great potential for global sharing from the local-community to the global optimization. Therefore, the cost of the proposed two-layer market, i.e., GS, is much lower than that of local-community optimization (LO).

The bidding curve $y_i(\varpi_i^0)$ of L-ESMs with 50, 100, and 200 prosumers with different elasticities a_i is shown in Fig. 7. A large a_i may lead to the saturation phenomenon, while a small a_i would decrease the price sensitivity of prosumers. Therefore, a moderate elasticity is necessary for energy sharing. Fig. 7 shows that the range of moderate elasticity is approximately inverse proportion to the scale of the L-ESM. Hence, the value of elasticity is supposed to reduce as the market scale increases. It indicates that the market clearing result will approximate to the social optimum in large-scale markets as discussed in Remark 2. Besides, Fig. 7 also verifies the monotonicity of $y_i(\varpi_i^0)$ disclosed in Proposition 5.

5.4. Sharing price without electric utility

The U-ESM outcome without the electric utility is shown in Fig. 8. It means that prosumers cannot trade with the electric utility, i.e., $p_j^+ = p_j^- = 0$. Without the competition of the electric utility, the energy sharing price is out of control. For the L-ESMs that purchases electricity through the U-ESM, the sharing prices sharply increase to almost 0.4 \$/kW, nearly twice of the buying price from the electric utility $\varpi^+ = 0.2$ \$/kW. Due to the line congestion, the sharing prices of L-ESMs 63–67 drop to lower than the selling price to the electric utility $\varpi^- = 0.05$ \$/kW. Though the shared energy in the U-ESM changes little, the sharing price fluctuates more intensely than that in Fig. 6. It verifies that the electric utility takes effect in limiting the market power exploitation in the sharing market.

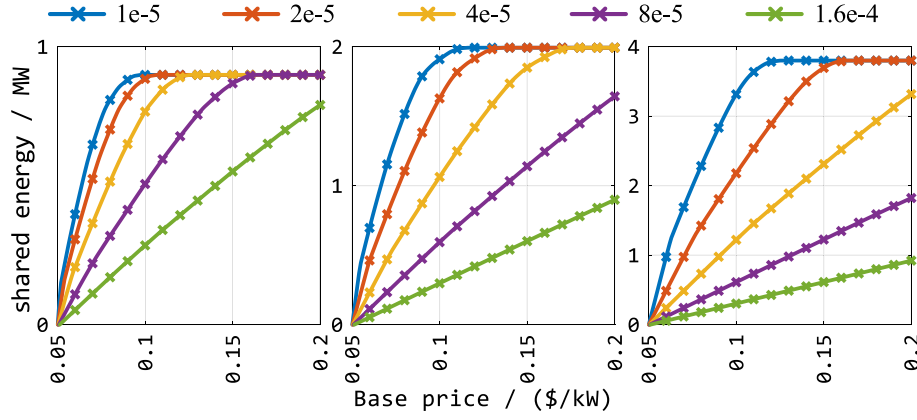


Fig. 7. Bidding curve $y_i(\omega_i^0)$ of L-ESMs with 50, 100, and 200 prosumers with different elasticities a_i .

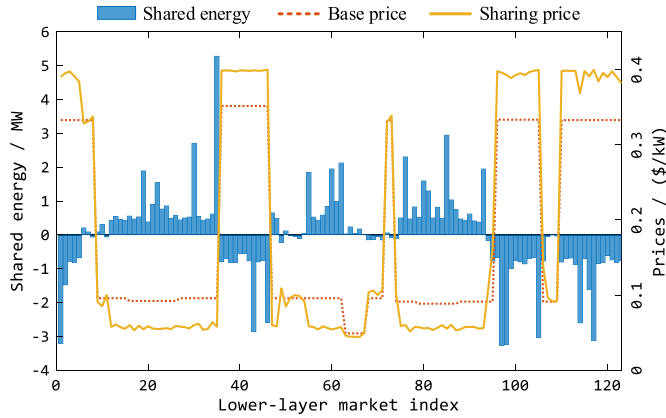


Fig. 8. Upper-layer energy sharing market outcome without the electric utility.

6. Conclusion

Wider-area sharing market can further promote the social efficiency than local-community ones by dealing with the uneven distribution of energy resources in local communities. This paper proposes a two-layer energy sharing market with an upper-layer energy sharing market and numerous lower-layer energy sharing markets. The lower-layer markets for prosumers in the same communities do not have to clear. The un-cleared energy of a lower-layer market is shared with other lower-layer markets in the upper-layer market. With the two-layer sharing, the social efficiency is further promoted. It is proved that the outcome of the proposed two-layer market approaches to the social optimum. Furthermore, to obtain the market outcome, this paper proposes an efficient hierarchically distributed bidding algorithm. Numerical experiments carried out on the IEEE 123-bus system with 11,250 prosumers demonstrate the social efficiency of the two-layer sharing market and the effectiveness of the hierarchically distributed bidding algorithm. It is expected that this study can inspire more studies on wider-area energy sharing for massive prosumers.

CRediT authorship contribution statement

Yifan Su: Writing – original draft, Validation, Software, Methodology. **Peng Yang:** Writing – review & editing, Visualization, Resources. **Kai Kang:** Writing – review & editing, Resources. **Zhaojian Wang:** Writing – review & editing, Methodology, Investigation. **Ning Qi:** Writing – review & editing. **Tonghua Liu:** Validation. **Feng Liu:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Proof of Proposition 1

Proof. Given $x_j, \forall j \in \mathcal{U}_i \setminus \{j\}$, assume that there is an optimal solution to (3) $s_j^* := \{p_j^*, p_j^{+*}, p_j^{-*}, x_j^*\}$ such that $p_j^{+*} > 0, p_j^{-*} > 0$. Then we can construct a strictly smaller solution than s_j^* .

Define $p_{j,\min} = \min \{p_j^{+*}, p_j^{-*}\} > 0$. Construct a new solution $s_j' := \{p_j^*, p_j^{+*} - p_{j,\min}, p_j^{-*} - p_{j,\min}, x_j^*\}$. It is easy to check that s_j' is feasible and that the purchased and sold electricity are complementary. Meanwhile, the objective value is smaller than the optimal value by $(\omega^+ - \omega^-)p_{j,\min} > 0$, which completes the proof by contradiction. $\square$

Appendix B. Proof of Proposition 2

Proof. We first prove that the NE is the optimal solution to problem (7) and then prove its existence and uniqueness.

(1) *Equivalent transformation.* Suppose there is an NE $s^{i*} := \{p_j^*, p_j^{+*}, p_j^{-*}, x_j^*, \forall j \in \mathcal{U}_i\}$. Since the NE is the optimal solution to (5), it satisfies the optimality condition [34, Thm. 3.24]

$$\langle \nabla F_j(s_j^*, s_{-j}^*), s_j - s_j^* \rangle \geq 0, \forall s_j \in S_j, j \in \mathcal{U}_i, \quad (\text{B.1})$$

where the gradient $\nabla F_j(s_j^*, s_{-j}^*)$ is defined as

$$\nabla F_j(s_j^*, s_{-j}^*) = \begin{bmatrix} c_j p_j^* + b_j \\ \omega^+ \\ -\omega^- \\ a_i x_j^* - \omega_i^0 + a_i \sum_{j' \in \mathcal{U}_i} x_{j'}^* \end{bmatrix} \quad (\text{B.2})$$

Note that the feasibility region of problem (7) is the Cartesian product of all action sets $S_j, \forall j \in \mathcal{U}_i$. It directly induces that the optimality condition of (7) is the collection of inequalities (B.1) for all $j \in \mathcal{U}_i$. Therefore, the NE of (5) is equivalent to the optimal solution to (7).

(2) *Existence of optimal solution.* Problem (7) is obviously feasible. If the objective value is always bounded, there must exist an optimal solution. From the weak duality [35, Sec. 5.2.2], the objective value of an arbitrary feasible solution to the dual problem provides a lower bound to the primal problem. Therefore, the existence of an optimal solution to problem (7) is equivalent to the feasibility of its dual problem. The constraints of the dual problem are given by

$$\varpi^+ - \mu_j^+ - \mu_j^{PB} = 0, \forall j \in \mathcal{U}_i \quad (\text{B.3})$$

$$-\varpi^- - \mu_j^- + \mu_j^{PB} = 0, \forall j \in \mathcal{U}_i \quad (\text{B.4})$$

$$\underline{\mu}_j, \bar{\mu}_j, \mu_j^+, \mu_j^-, \mu_j^{PB} \geq 0, \forall j \in \mathcal{U}_i. \quad (\text{B.5})$$

We can construct a feasible solution as $\underline{\mu}_j = \bar{\mu}_j = \mu_j^+ = 0, \mu_j^- = \varpi^+ - \varpi^-, \mu_j^{PB} = \varpi^+, \forall j \in \mathcal{U}_i$, which proves the existence of optimal solution.

(3) *Uniqueness of optimal solution.* Assume that there are two different optimal solutions to problem (7) denoted by $\hat{s}^i := \{\hat{p}_j, \hat{p}_j^+, \hat{p}_j^-, \hat{x}_j, \forall j \in \mathcal{U}_i\}$ and $\check{s}^i := \{\check{p}_j, \check{p}_j^+, \check{p}_j^-, \check{x}_j, \forall j \in \mathcal{U}_i\}$. Suppose that the two solutions differ in some decisions of prosumers indexed by $j \in \mathcal{U}^d \neq \emptyset$. By Proposition 1, at the NE, given $\{p_j, x_j\}$, there exist a unique and defined $\{p_j^+, p_j^-\}$. Thus for each prosumer $j \in \mathcal{U}^d$, their $\{p_j, x_j\}$ must be not identical, i.e., $\{\hat{p}_j, \hat{x}_j\} \neq \{\check{p}_j, \check{x}_j\}, \forall j \in \mathcal{U}^d$.

Divide the objective of problem (7) into two parts: 1) a strongly convex function $F^{(1)}(\mathbf{p}^i, \mathbf{x}^i)$ and 2) a linear function $F^{(2)}(\mathbf{p}^{i+}, \mathbf{p}^{i-})$. Let F^* denote the optimal value of problem (7), i.e., $F^* = F^{(1)}(\hat{\mathbf{p}}^i, \hat{\mathbf{x}}^i) + F^{(2)}(\hat{\mathbf{p}}^{i+}, \hat{\mathbf{p}}^{i-}) = F^{(1)}(\check{\mathbf{p}}^i, \check{\mathbf{x}}^i) + F^{(2)}(\check{\mathbf{p}}^{i+}, \check{\mathbf{p}}^{i-})$.

We can construct another solution $\tilde{s}^i = (\hat{s}^i + \check{s}^i)/2$, which is obviously feasible. Since the optimal solutions $\hat{s}^i$ and $\check{s}^i$ differ in $\{\mathbf{p}^i, \mathbf{x}^i\}$, by the strong convexity of $F^{(1)}(\mathbf{p}^i, \mathbf{x}^i)$ and the linearity of $F^{(2)}(\mathbf{p}^{i+}, \mathbf{p}^{i-})$, we have

$$\begin{aligned} & F^{(1)}(\tilde{\mathbf{p}}^i, \tilde{\mathbf{x}}^i) + F^{(2)}(\tilde{\mathbf{p}}^{i+}, \tilde{\mathbf{p}}^{i-}) \\ & < (F^{(1)}(\hat{\mathbf{p}}^i, \hat{\mathbf{x}}^i) + F^{(1)}(\check{\mathbf{p}}^i, \check{\mathbf{x}}^i))/2 \\ & \quad + (F^{(2)}(\hat{\mathbf{p}}^{i+}, \hat{\mathbf{p}}^{i-}) + F^{(2)}(\check{\mathbf{p}}^{i+}, \check{\mathbf{p}}^{i-}))/2 \\ & = F^* \end{aligned} \quad (\text{B.6})$$

It means that if there are non-unique optimal solutions to problem (7), we can always find another solution with a smaller objective value than the optimal value. The uniqueness of the optimal solution is proved by contradiction, which completes the proof. $\square$

Appendix C. Proof of Proposition 4

Proof. For each prosumer $j \in \mathcal{U}_i$, given $\mathbf{x}_j^*$, the NE satisfies the KKT condition of problem (3)

$$c_j p_j^* + b_j - \underline{\mu}_j^* + \bar{\mu}_j^* - \mu_j^{PB*} = 0 \quad (\text{C.1})$$

$$\varpi^+ - \mu_j^{+*} - \mu_j^{PB*} = 0 \quad (\text{C.2})$$

$$-\varpi^- - \mu_j^{-*} + \mu_j^{PB*} = 0 \quad (\text{C.3})$$

$$-\varpi_i^0 + a_i x_j^* + a_i \sum_{j' \in \mathcal{U}_i} x_{j'}^* + \mu_j^{PB*} = 0 \quad (\text{C.4})$$

$$0 \leq p_j^* - \underline{p}_j \perp \underline{\mu}_j^* \geq 0 \quad (\text{C.5})$$

$$0 \leq \bar{p}_j - p_j^* \perp \bar{\mu}_j^* \geq 0 \quad (\text{C.6})$$

$$0 \leq p_j^{+*} \perp \mu_j^{+*} \geq 0 \quad (\text{C.7})$$

$$0 \leq p_j^{-*} \perp \mu_j^{-*} \geq 0 \quad (\text{C.8})$$

$$D_j + x_j^* + p_j^{+*} = p_j^* + p_j^{+*} \quad (\text{C.9})$$

(a) Combining (C.2) and (C.7), we have

$$\varpi^+ - \mu_j^{PB*} = \mu_j^{+*} \geq 0.$$

Similarly, the combination of (C.3) and (C.8) yields

$$\mu_j^{PB*} - \varpi^- = \mu_j^{-*} \geq 0.$$

(b) Recalling the definition of $\varpi_i(\mathbf{x}^{i*})$, Eq. (C.4) directly yields

$$x_j^* = \frac{\varpi_i(\mathbf{x}^{i*}) - \mu_j^{PB*}}{a_i}.$$

(c) Summing (C.4) over all $j \in \mathcal{U}_i$, we have

$$-|\mathcal{U}_i| \varpi_i^0 + a_i(1 + |\mathcal{U}_i|) \sum_{j \in \mathcal{U}_i} x_j^* + \sum_{j \in \mathcal{U}_i} \mu_j^{PB*} = 0$$

By the definition of $\varpi_i(\mathbf{x}^{i*})$, Eq. (13) holds directly.

(d) By (13), the value of the energy sharing price $\varpi_i(\mathbf{x}^{i*})$ is the average of the base price ϖ_i^0 and shadow prices of prosumers. Recall that $\mu_j^{PB*}, \forall j \in \mathcal{U}_i$ are in the range $[\varpi^-, \varpi^+]$. If ϖ_i^0 is also bounded by ϖ^+ and ϖ^- , their average is obviously located in the same range.

As the scale of prosumers increases, we have

$$\begin{aligned} \lim_{|\mathcal{U}_i| \rightarrow \infty} \varpi_i(\mathbf{x}^{i*}) &= \lim_{|\mathcal{U}_i| \rightarrow \infty} \frac{\varpi_i^0 + \sum_{j \in \mathcal{U}_i} \mu_j^{PB*}}{1 + |\mathcal{U}_i|} \\ &= \lim_{|\mathcal{U}_i| \rightarrow \infty} \frac{\sum_{j \in \mathcal{U}_i} \mu_j^{PB*}}{|\mathcal{U}_i|}. \end{aligned}$$

Even if ϖ_i^0 is out of $[\varpi^-, \varpi^+]$, the energy sharing price also approaches the range, which completes the proof. $\square$

Appendix D. Proof of Proposition 5

Proof. We will prove it by contradiction. Assume that $y_i(\varpi_i^0)$ is monotonically decreasing with respect to ϖ_i^0 . At the NE, a larger ϖ_i^0 causes a smaller $y_i(\varpi_i^0)$. Then by the definition (4), the energy sharing price $\varpi_i(\mathbf{x}^{i*})$ increases. Then discuss by the trend of μ_j^{PB*} .

(a) μ_j^{PB*} **decreases.** By (C.1), (C.5) and (C.6), the power generation is given by

$$p_j^* = \left[\frac{\mu_j^{PB*} - b_j}{c_j} \right]_{\underline{p}_j}^{\bar{p}_j}, \quad (\text{D.1})$$

where $[\cdot]_a^b$ is the projection onto the range $[a, b]$. Thus p_j^* decreases or remains. By (C.2) and (C.7), if $\mu_j^{PB*} \in [\varpi^-, \varpi^+]$, we have $\mu_j^{+*} > 0$, which indicates $p_j^{+*} = 0$. Only if $\mu_j^{PB*} = \varpi^+$, p_j^{+*} may be positive. Therefore, in this situation, p_j^{+*} decreases or remains zero. Similarly, p_j^{-*} may be positive if $\mu_j^{-*} = \varpi^-$; otherwise, it remains zero. Thus p_j^{-*} increases or remains zero in this situation. Besides, (C.4) indicates that x_j^* must increase. However, the trends of $p_j^*, p_j^{+*}, p_j^{-*}$ and x_j^* do not satisfy the power balance constraint (C.9). It means that the situation of decreasing μ_j^{PB*} is impossible.

(b) μ_j^{PB*} **increases.** Similarly to a), an increasing μ_j^{PB*} can deduce that p_j^* and p_j^{+*} increase or remain, while p_j^{-*} decreases or remains zero. By the power balance constraint (C.9), x_j^* will increase or remain, which does not violate (C.4). Therefore, in the situation of increasing μ_j^{PB*} , x_j^* will increase or remain.

(c) μ_j^{PB*} **remains.** By (C.4), x_j^* must increase. There is an possible condition, where $\mu_j^{PB*} \equiv \varpi^+$ and p_j^{+*} increases, while p_j^* and p_j^{-*} remain.

To sum up, except for the impossible situation of decreasing μ_j^{PB*} , the other two situations can deduce that x_j^* will increase or remain. Summing over $\forall j \in \mathcal{U}_i$, $y_i(\varpi_i^0) = \sum_{j \in \mathcal{U}_i} x_j^*$ also increases or remains. It does not match the initial assumption of decreasing $y_i(\varpi_i^0)$, which proves the original proposition by contradiction. $\square$

Appendix E. Proof of Proposition 6

Proof. We first prove that Algorithm 2 is equivalent to the dual decomposition method to solve the problem (21). Following [3,36], define the Lagrangian of problem (21)

$$\begin{aligned} L = & \sum_{i \in \mathcal{N}} \sum_{j \in \mathcal{U}_i} \left\{ \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- \right\} \\ & + \sum_{i \in \mathcal{N}} \left\{ \frac{a_i}{2} y_i^2 + \frac{a_i}{2} \sum_{j \in \mathcal{U}_i} x_j^2 \right\} \\ & + \lambda^{PB} \sum_{i \in \mathcal{N}} y_i + \sum_{l \in \mathcal{L}} \lambda_l \left(\sum_{i \in \mathcal{N}} \pi_{il} y_i - F_l \right) \end{aligned} \quad (\text{E.1})$$

where λ^{PB} and λ_l are the Lagrangian multiplier of (21d) and (21e), respectively.

The dual decomposition method is alternately solving subproblems for $\forall i \in \mathcal{N}$

$$\begin{aligned} \min \sum_{j \in \mathcal{U}_i} & \left\{ \frac{c_j}{2} p_j^2 + b_j p_j + \varpi^+ p_j^+ - \varpi^- p_j^- \right\} + \frac{a_i}{2} \sum_{j \in \mathcal{U}_i} x_j^2 \\ & + \frac{a_i}{2} y_i^2 + (\lambda^{PB}(k) + \sum_{l \in \mathcal{L}} \pi_{il} \lambda_l(k)) y_i \end{aligned} \quad (\text{E.2a})$$

$$\text{s.t. } \{p_j, p_j^+, p_j^-, x_j\} \in S_j, \forall j \in \mathcal{U}_i \quad (\text{E.2b})$$

$$y_i = \sum_{j \in \mathcal{U}_i} x_j \quad (\text{E.2c})$$

to attain $y_i(k)$, $\forall i \in \mathcal{N}$ and update Lagrangian multipliers

$$\lambda^{PB}(k+1) = \lambda^{PB}(k) + \alpha^{PB} \sum_{i \in \mathcal{N}} y_i(k) \quad (\text{E.3a})$$

$$\lambda_l(k+1) = \left[\lambda_l(k) + \alpha_l \left(\sum_{i \in \mathcal{N}} \pi_{il} y_i(k) - F_l \right) \right]^+ \quad (\text{E.3b})$$

where $[\cdot]^+$ is the projection onto the range $[0, +\infty)$.

Let $\lambda^{PB} \equiv -\varpi^{PB}$ and $\lambda_l \equiv -\varpi_l$. Then updating Lagrangian multipliers (E.3) is equivalent to adjusting prices (20). Meanwhile, solving subproblem (E.2) equals to seeking the NE of the L-ESM according to Proposition 2. Therefore, Algorithm 2 converges to the optimal solution to problem (21).

The existence and uniqueness of the optimal solution to problem (21) are just the same as those of problem (7) in Proposition 2. Due to the space limitation, the proof is omitted here. $\square$

Data availability

Data will be made available upon request.

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